

# Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

## Practise-1

2520030098

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Section – 3

Create the database:

```
CREATE DATABASE IF NOT EXISTS bookflow5_db;
```

Output:

```
✓ 143 09:12:06 CREATE DATABASE IF NOT EXISTS bookflow5_db
```

Use database:

```
USE bookflow5_db;
```

Output:

```
✓ 145 09:14:03 USE bookflow5_db
```

Create Physician Table:

```
CREATE TABLE physician (  
    employeeid INT PRIMARY KEY,  
    name VARCHAR(100),  
    position VARCHAR(100),  
    ssn VARCHAR(20)  
);
```

Output:

```
✓ 146 09:15:11 CREATE TABLE physician ( employeeid INT PRIMARY KEY, name VARCHAR(100), position VA
```

Insert Data Into Physician:

```

INSERT INTO physician VALUES
(1,'Sushma','Staff Internist','11111111'),
(2,'Sowmya','Attending Physician','22222222'),
(3,'Mokshagna','Surgical Attending Physician','33333333'),
(4,'Akhila','Senior Attending Physician','44444444'),
(5,'Likhitha','Head Chief of Medicine','55555555'),
(6,'Vaishnavi','Surgical Attending Physician','66666666'),
(7,'Manasvi','Surgical Attending Physician','77777777'),
(8,'Sairam','MD Resident','88888888'),
(9,'Krishna','Attending Psychiatrist','99999999');

```

Output:

147 09:20:16 INSERT INTO physician VALUES (1,'Sushma','Staff Internist','11111111'), (2,'Sowmya','Attending Physician',

Select:

```

SELECT * FROM physician;

```

Output:

|   | employeeid | name      | position                     | ssn      |
|---|------------|-----------|------------------------------|----------|
| ▶ | 1          | Sushma    | Staff Internist              | 11111111 |
|   | 2          | Sowmya    | Attending Physician          | 22222222 |
|   | 3          | Mokshagna | Surgical Attending Physician | 33333333 |
|   | 4          | Akhila    | Senior Attending Physician   | 44444444 |
|   | 5          | Likhitha  | Head Chief of Medicine       | 55555555 |
|   | 6          | Vaishnavi | Surgical Attending Physician | 66666666 |
|   | 7          | Manasvi   | Surgical Attending Physician | 77777777 |
|   | 8          | Sairam    | MD Resident                  | 88888888 |
|   | 9          | Krishna   | Attending Psychiatrist       | 99999999 |
| * | NULL       | NULL      | NULL                         | NULL     |

Create Department Table:

```

CREATE TABLE department (
    departmentid INT PRIMARY KEY,
    name VARCHAR(100),
    head INT
);

```

Output:

149 09:23:25 CREATE TABLE department ( departmentid INT PRIMARY KEY, name VARCHAR(100), head INT )

Insert Data Into Department:

```

INSERT INTO department VALUES
(1,'General Medicine',4),
(2,'Surgery',7),
(3,'Psychiatry',9);
SELECT * FROM department;

```

Output:

|   | departmentid | name             | head |
|---|--------------|------------------|------|
| ▶ | 1            | General Medicine | 4    |
|   | 2            | Surgery          | 7    |
|   | 3            | Psychiatry       | 9    |
| * | NULL         | NULL             | NULL |

Inner Join:

```

SELECT d.name AS Department,
       p.name AS Head_Physician
FROM department d
JOIN physician p
ON d.head = p.employeeid;

```

Output:

|   | Department       | Head_Physician |
|---|------------------|----------------|
| ▶ | General Medicine | Akhila         |
|   | Surgery          | Manasvi        |
|   | Psychiatry       | Krishna        |

Where Clause:

```

SELECT *
FROM physician
WHERE position='Surgical Attending Physician';

```

Output:

|   | employeeid | name      | position                     | ssn       |
|---|------------|-----------|------------------------------|-----------|
| ▶ | 3          | Mokshagna | Surgical Attending Physician | 333333333 |
|   | 6          | Vaishnavi | Surgical Attending Physician | 666666666 |
|   | 7          | Manasvi   | Surgical Attending Physician | 777777777 |
| * | NULL       | NULL      | NULL                         | NULL      |

Where + Like:

```

SELECT *
FROM physician
WHERE name LIKE 'Sushma%';

```

Output:

|   | employeeid | name   | position        | ssn       |
|---|------------|--------|-----------------|-----------|
| ▶ | 1          | Sushma | Staff Internist | 111111111 |
| * | NULL       | NULL   | NULL            | NULL      |

Count():

```
SELECT COUNT(*) AS Total_Physicians  
FROM physician;
```

Output:

|   | Total_Physicians |
|---|------------------|
| ▶ | 9                |

Count Distinct:

```
SELECT COUNT(DISTINCT position)  
FROM physician
```

Output:

|   | COUNT(DISTINCT<br>position) |
|---|-----------------------------|
| ▶ | 7                           |

Group By:

```
SELECT position,  
       COUNT(*) AS Total_Employees  
FROM physician  
GROUP BY position;
```

Output:

|   | position                     | Total_Employees |
|---|------------------------------|-----------------|
| ▶ | Staff Internist              | 1               |
|   | Attending Physician          | 1               |
|   | Surgical Attending Physician | 3               |
|   | Senior Attending Physician   | 1               |
|   | Head Chief of Medicine       | 1               |
|   | MD Resident                  | 1               |
|   | Attending Psychiatrist       | 1               |

Group By + Having:

```
SELECT position,
        COUNT(*) AS Total_Employees
FROM physician
GROUP BY position
HAVING COUNT(*) > 1;
```

Output:

|   | position                     | Total_Employees |
|---|------------------------------|-----------------|
| ▶ | Surgical Attending Physician | 3               |

Order By:

```
SELECT *
FROM physician
ORDER BY name ASC;
```

Output:

|   | employeeid | name      | position                     | ssn       |
|---|------------|-----------|------------------------------|-----------|
| ▶ | 4          | Akhila    | Senior Attending Physician   | 444444444 |
|   | 9          | Krishna   | Attending Psychiatrist       | 999999999 |
|   | 5          | Likhitha  | Head Chief of Medicine       | 555555555 |
|   | 7          | Manasvi   | Surgical Attending Physician | 777777777 |
|   | 3          | Mokshagna | Surgical Attending Physician | 333333333 |
|   | 8          | Sairam    | MD Resident                  | 888888888 |
|   | 2          | Sowmya    | Attending Physician          | 222222222 |
|   | 1          | Sushma    | Staff Internist              | 111111111 |
|   | 6          | Vaishnavi | Surgical Attending Physician | 666666666 |
| ★ | NULL       | NULL      | NULL                         | NULL      |

Order By Desc:

```
SELECT *
FROM physician
ORDER BY employeeid DESC;
```

Output:

|   | employeeid | name      | position                     | ssn       |
|---|------------|-----------|------------------------------|-----------|
| ▶ | 9          | Krishna   | Attending Psychiatrist       | 999999999 |
|   | 8          | Sairam    | MD Resident                  | 888888888 |
|   | 7          | Manasvi   | Surgical Attending Physician | 777777777 |
|   | 6          | Vaishnavi | Surgical Attending Physician | 666666666 |
|   | 5          | Likhitha  | Head Chief of Medicine       | 555555555 |
|   | 4          | Akhila    | Senior Attending Physician   | 444444444 |
|   | 3          | Mokshagna | Surgical Attending Physician | 333333333 |
|   | 2          | Sowmya    | Attending Physician          | 222222222 |
|   | 1          | Sushma    | Staff Internist              | 111111111 |
| ★ | NULL       | NULL      | NULL                         | NULL      |

## Where + In:

```
SELECT *  
FROM physician  
WHERE employeeid IN  
(  
    SELECT head  
    FROM department  
);
```

## Output:

|   | employeeid | name    | position                     | ssn       |
|---|------------|---------|------------------------------|-----------|
| ▶ | 4          | Akhila  | Senior Attending Physician   | 444444444 |
|   | 7          | Manasvi | Surgical Attending Physician | 777777777 |
|   | 9          | Krishna | Attending Psychiatrist       | 999999999 |
| * | NULL       | NULL    | NULL                         | NULL      |

## Subquery:

```
SELECT *  
FROM physician  
WHERE employeeid NOT IN  
(  
    SELECT head  
    FROM department  
);
```

## Output:

|   | employeeid | name      | position                     | ssn       |
|---|------------|-----------|------------------------------|-----------|
| ▶ | 1          | Sushma    | Staff Internist              | 111111111 |
|   | 2          | Sowmya    | Attending Physician          | 222222222 |
|   | 3          | Mokshagna | Surgical Attending Physician | 333333333 |
|   | 5          | Likhitha  | Head Chief of Medicine       | 555555555 |
|   | 6          | Vaishnavi | Surgical Attending Physician | 666666666 |
|   | 8          | Sairam    | MD Resident                  | 888888888 |
| * | NULL       | NULL      | NULL                         | NULL      |

## Group By + Order By:

```
SELECT position,  
       COUNT(*) AS Total_Employees  
FROM physician  
GROUP BY position  
ORDER BY Total_Employees DESC;
```

## Output:

|   | position                     | Total_Employees |
|---|------------------------------|-----------------|
| ▶ | Surgical Attending Physician | 3               |
|   | Staff Internist              | 1               |
|   | Attending Physician          | 1               |
|   | Senior Attending Physician   | 1               |
|   | Head Chief of Medicine       | 1               |
|   | MD Resident                  | 1               |
|   | Attending Psychiatrist       | 1               |

## Where + Like

```
SELECT *
FROM physician
WHERE position LIKE '%Attending%';
```

## Output:

|   | employeeid | name      | position                     | ssn       |
|---|------------|-----------|------------------------------|-----------|
| ▶ | 2          | Sowmya    | Attending Physician          | 222222222 |
|   | 3          | Mokshagna | Surgical Attending Physician | 333333333 |
|   | 4          | Akhila    | Senior Attending Physician   | 444444444 |
|   | 6          | Vaishnavi | Surgical Attending Physician | 666666666 |
|   | 7          | Manasvi   | Surgical Attending Physician | 777777777 |
|   | 9          | Krishna   | Attending Psychiatrist       | 999999999 |
| * | NULL       | NULL      | NULL                         | NULL      |

## Join + Where

```
SELECT p.name
FROM physician p
JOIN department d
ON p.employeeid=d.head
WHERE d.name='Surgery';
```

## Output:

|   | name    |
|---|---------|
| ▶ | Manasvi |

## Join + Order By

```
SELECT d.name,
       p.name
FROM department d
JOIN physician p
ON d.head=p.employeeid
ORDER BY d.name;
```

## Output:



|   | name             | name    |
|---|------------------|---------|
| ► | General Medicine | Akhila  |
|   | Psychiatry       | Krishna |
|   | Surgery          | Manasvi |

## Where + Not Like

```
SELECT *
FROM physician
WHERE position NOT LIKE '%Surgical%';
```

## Output:

|   | employeeid | name     | position                   | ssn       |
|---|------------|----------|----------------------------|-----------|
| ► | 1          | Sushma   | Staff Internist            | 111111111 |
|   | 2          | Sowmya   | Attending Physician        | 222222222 |
|   | 4          | Akhila   | Senior Attending Physician | 444444444 |
|   | 5          | Likhitha | Head Chief of Medicine     | 555555555 |
|   | 8          | Sairam   | MD Resident                | 888888888 |
|   | 9          | Krishna  | Attending Psychiatrist     | 999999999 |
| * | NULL       | NULL     | NULL                       | NULL      |

## Count + Having

```
SELECT position,
       COUNT(*) AS Total
FROM physician
GROUP BY position
HAVING COUNT(*) >= 2;
```

## Output:

|   | position                     | Total |
|---|------------------------------|-------|
| ► | Surgical Attending Physician | 3     |

## Exists

```
SELECT *
FROM department d
WHERE EXISTS
(
  SELECT 1
  FROM physician p
  WHERE p.employeeid=d.head
);
```

## Output:



|   | departmentid | name             | head |
|---|--------------|------------------|------|
| ▶ | 1            | General Medicine | 4    |
|   | 2            | Surgery          | 7    |
|   | 3            | Psychiatry       | 9    |
| • | NULL         | NULL             | NULL |

Max()

```
SELECT MAX(employeeid)
FROM physician;
```

Output:

|   | MAX(employeeid) |
|---|-----------------|
| ▶ | 9               |

Min():

```
SELECT MIN(employeeid)
FROM physician;
```

Output:

|   | MIN(employeeid) |
|---|-----------------|
| ▶ | 1               |